Four fresh blind targets; all 12 material values unknown; 3 s per method
Top1 uses all-data residual. Oracle column uses GT only after predictions froze.

HOLDOUT12_00

Ground truth
2.6 23.3 23.3 23.3
11.7 11.7 23.3 2.6
2.6 11.7 2.6 2.6

FullJ top1
RMS 10.8 | rel. residual 0.068
2.3 20.1 0.5 25.7
13.0 9.1 16.3 2.1
30.0 11.3 10.1 0.8

GlobalDual top1
RMS 6.33e-09 | rel. residual 5.2e-11
2.6 23.3 23.3 23.3
11.7 11.7 23.3 2.6
2.6 11.7 2.6 2.6

GlobalDual oracle best of 22
RMS 6.33e-09 | GT selected
2.6 23.3 23.3 23.3
11.7 11.7 23.3 2.6
2.6 11.7 2.6 2.6

HOLDOUT12_01

Ground truth
10.3 1.9 1.9 1.9
10.3 10.3 1.9 1.9
10.3 10.3 1.9 1.9

FullJ top1
RMS 1.26e-11 | rel. residual 2.1e-13
10.3 1.9 1.9 1.9
10.3 10.3 1.9 1.9
10.3 10.3 1.9 1.9

GlobalDual top1
RMS 8.32 | rel. residual 0.25
6.9 2.0 4.6 2.8
11.8 10.4 30.0 0.5
11.0 7.3 2.6 4.1

GlobalDual oracle best of 221
RMS 8.26 | GT selected
10.0 30.0 6.0 0.5
9.0 10.4 0.5 1.2
11.4 11.5 0.5 2.6

HOLDOUT12_02

Ground truth
2.4 13.8 13.8 13.8
2.4 2.4 13.8 21.7
2.4 2.4 21.7 21.7

FullJ top1
RMS 10.8 | rel. residual 0.13
30.0 3.4 8.8 16.0
0.5 12.3 19.1 18.9
8.3 19.8 17.0 18.5

GlobalDual top1
RMS 8.68 | rel. residual 0.059
3.3 13.9 13.1 13.8
0.5 8.7 13.1 20.2
7.5 30.0 30.0 20.1

GlobalDual oracle best of 255
RMS 3.96 | GT selected
10.5 13.5 20.4 9.5
1.5 0.5 15.7 17.8
0.5 6.8 17.8 20.9

HOLDOUT12_03

Ground truth
14.3 14.3 14.3 2.1
2.1 2.1 2.1 21.3
2.1 2.1 21.3 21.3

FullJ top1
RMS 10 | rel. residual 0.046
13.4 13.5 11.9 7.2
2.6 8.0 7.6 21.8
7.3 26.9 0.5 15.9

GlobalDual top1
RMS 1.52e-13 | rel. residual 1.1e-15
14.3 14.3 14.3 2.1
2.1 2.1 2.1 21.3
2.1 2.1 21.3 21.3

GlobalDual oracle best of 97
RMS 1.52e-13 | GT selected
14.3 14.3 14.3 2.1
2.1 2.1 2.1 21.3
2.1 2.1 21.3 21.3

Material value (public box 0.5–30)
30
25
20
15
10
5